

Visual Generative Artificial Intelligence (GenAI) in Mixed Reality (MR): A Survey of Techniques, Interaction Designs, and Evaluations

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Abstract—GenAI is increasingly reshaping spatial interactions in immersive environments, bridging diverse input and output modalities to create intelligent user experiences. This survey provides a structured examination of visual GenAI techniques adapted and applied to MR contexts, which include Virtual Reality (VR), and Augmented Reality (AR), emphasizing their roles in enabling rich, adaptive, and naturalistic interactions. We screened publications from international venues between 2018 and 2025, followed by an in-depth review of 102 papers. We focus on the applications of visual content creation in 2D&3D and vision-language models (VLMs), highlighting their integration practices within MR. Furthermore, we investigate the co-evolution of generative models and MR interaction design, including immersive visualization, controllable generation, and synthetic data creation. The survey concludes by outlining recent challenges and potential research directions to advance the synergy of visual GenAI and MR techniques.

Keywords—Generative model, virtual reality, augmented reality, human-AI collaboration, scoping review

I. INTRODUCTION

Immersive environments (VR/AR) [1]–[3] are increasingly adopted across design, education, and entertainment. However, content creation within these spaces remains a significant bottleneck. Traditional tools [4]–[6] rely on labor-intensive, rule-based processes that require specialized expertise, hindering widespread innovation. GenAI has emerged as a transformative solution, automating the creation and refinement of visual assets. By leveraging breakthroughs in diffusion models [7], autoregressive structures [8], and vision-language embeddings [9]–[11], GenAI can synthesize and reason about high-quality content with sophisticated multimodal pipelines and large datasets [12]–[17]. This includes diverse outputs—ranging from images [18] and videos [19] to 3D models [20], [21]. In this survey, **“visual GenAI”** refers to generative models whose inputs, outputs, or intermediate representations involve 2D images or 3D objects, and that enable the creation, manipulation, or adaptation of such visual content. We also include GenAI systems that

orchestrate these processes through LLMs, VLMs, or multimodal reasoning models.

Despite parallel advancements in MR and GenAI, a critical gap exists in their integration. While AI techniques are well-established in MR for perception and interaction, enhancing tasks such as gesture recognition [22], [23], depth estimation [24], and layout optimization [25], the role of visual GenAI within these environments remains underexplored. This disconnect represents a major opportunity. Unlike desktop interfaces, which are limited to keyboards and mice, MR offers rich spatial interactions [26]–[28] that could fundamentally reshape how users control generative models. Early systems [10], [29], [30] have demonstrated this potential, enabling users to generate and arrange 3D assets via natural language commands (e.g., “create a kitchen”). However, systematic analyses of the interaction frameworks, technical challenges, and design considerations for GenAI in MR remain scarce [31], underscoring the need for this investigation.

Current literature remains segmented, with GenAI and MR research largely developing in parallel rather than interacting. Reviews of generative frameworks [7], [32]–[37] and HCI applications (e.g., co-creation [38]–[40], visual search [41]) predominantly focus on desktop environments, neglecting the unique spatial interaction paradigms of MR. Conversely, surveys anchored in MR [31], [42]–[45] often treat GenAI as peripheral or limit its scope to text-based LLMs [46], failing to demonstrate the transformative potential of visual generation (Table I). Hirzle et al. [31] provide a broad scoping review of AI in XR, establishing an important foundation for understanding how learning-based techniques support perception, interaction, and system adaptation in immersive environments. However, their review largely predates the recent wave of LLMs, VLMs, and 2D/3D generative pipelines that now shape GenAI-enabled MR. Ning et al. [45] systematically review GenAI in XR, but their synthesis is organized primarily by general application domains, without in-depth discussions on evaluations and MR for GenAI. Even recent “Metaverse” surveys [47]–[50], which outline broad visions for immersive world creation, tend to be high-level; they largely overlook how specific interaction modalities (e.g., gaze and hand tracking) integrate with generative workflows. This fragmented landscape highlights the absence of a comprehensive review

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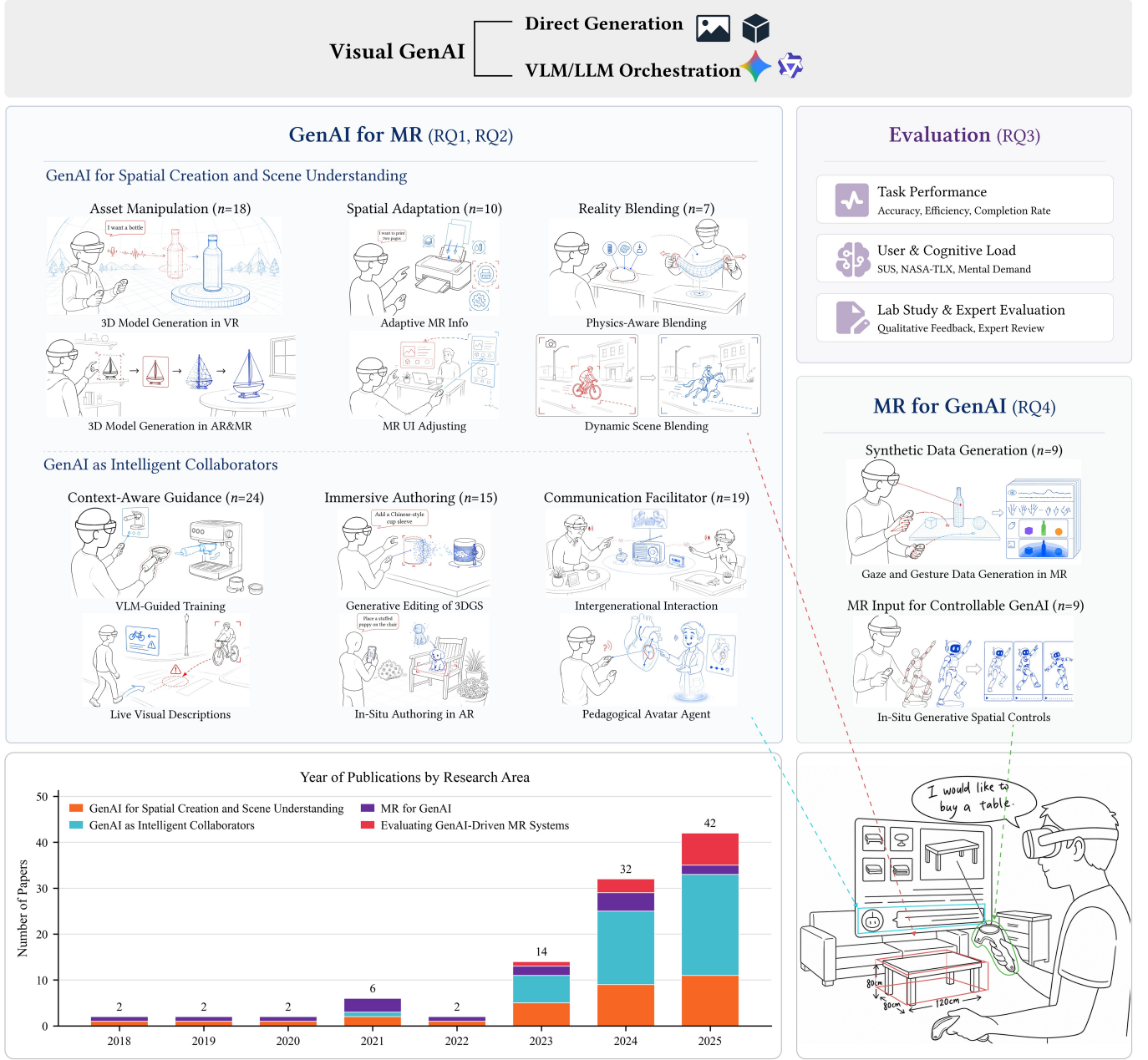


Fig. 1. Landscape of visual GenAI in MR. It summarizes 102 research works across the proposed research questions, while the temporal chart shows publication trends across research areas. n denotes the number of papers organized and reviewed for each theme. The lower-right panel presents a conceptual furniture-design workflow linked to three topics covered by our taxonomy: (1) GenAI as Intelligent Collaborators, where an in-situ GenAI agent interprets the user's request and recommends candidate furniture (cyan dashed line); (2) Reality Blending, where the selected table is placed and generative edited within the physical environment (red dashed line); and (3) MR Input for Controllable GenAI, where spatial controls allow the user to adjust the table's dimensions and iteratively refine the design (green dashed line).

examining the reciprocal development of Visual GenAI and MR spatial computing.

This paper addresses this gap by systematically exploring visual GenAI as part of the MR interaction loop: how MR users specify intent through speech, gaze, gesture, sketching, and scene capture; how generative models transform those inputs into spatial content or context-aware assistance; how MR platforms can advance GenAI development in turn; and how such systems are evaluated for usability, generation quality, workload, and user experience (Figure 1). Following Preferred

Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [51] and scoping review process suggested by Cooper et al. [52], we selected and analyzed 102 articles from top-tier venues published between 2018 and 2025 (Figure 3), as recognized by Google Scholar Metrics [53]. Our review synthesizes this body of work to answer four key research questions (RQs):

- **RQ1:** How does GenAI enhance spatial interaction and dynamic scene understanding in MR?
- **RQ2:** How have GenAI-powered agents been deployed

TABLE I
POSITIONING OUR SURVEY RELATIVE TO PRIOR REVIEWS ON XR, AI, GENAI, AND INTERACTION.

Survey	Primary Scope	Reporting Framework	Coverage	Taxonomy Organization
Hirzle et al. [31]	Broad XR-AI map, mutual development of AI techniques and XR systems; no GenAI focus.	Scoping review with PRISMA-guided selection and reporting.	311 full papers from 2017–2021 across 203 publication venues.	Content creation, user understanding, interaction support, intelligent agents, and XR for AI.
Hashim et al. [42]	GenAI for VR, with emphasis on content generation and user interactions.	Systematic review that follows the identification, screening, and eligibility process.	26 articles from 2020–2024, retrieved mainly from Scopus and JSTOR.	Immersive environments, content generation, and interaction types.
Mendes et al. [43]	3D object manipulation across desktop, touch, and mid-air interfaces; no GenAI focus.	Narrative survey that classifies manipulation techniques of object spatial transformations.	35 papers between 1987 and 2016, retrieved mainly from I3D, CHI, UIST, VRST, and IEEE VR.	Display/input properties and mappings for translation, rotation, and scaling.
Wang et al. [44]	Multimodal interaction for XR headsets; AI/LLMs mainly as interaction enablers.	Systematic review that follows the identification, screening, and eligibility process.	104 papers from 2022–2024 in CHI, UIST, IMWUT, IEEE VR, ISMAR, TVCG, and arXiv.	Applications, operation types, performance measures, and interaction modalities.
Ning et al. [45]	Closest prior review: GenAI applications in XR interactions and immersive techniques.	Systematic review with PRISMA-guided selection and reporting.	26 articles from 2023–2025 in ACM, IEEE, Google Scholar, and ScienceDirect.	Applications, generative models, user input, and feedback cues.
Tang et al. [46]	LLM integration in XR through awareness expansion; LLM-centered applications.	Scoping review that follows a Grounded Theory.	135 papers from 2021–2024 in ACM DL, Scopus, Web of Science, and arXiv.	Application domains, awareness types, LLM effects, evaluation metrics, and challenges.
Ours	Focusing on situated generation, multimodal control, interaction design, evaluation practice, and MR-supported GenAI.	Scoping review following Cooper et al. [52]; PRISMA-guided selection and reporting.	102 papers from 2018–2025 in ACM DL, IEEE Xplore, ScienceDirect, SpringerLink, Google Scholar, and arXiv.	Multi-label taxonomy coding: spatial adaptation, reality blending, immersive authoring, context-aware guidance, and MR for GenAI.

as intelligent collaborators in MR?

- **RQ3:** How have user experiences of these GenAI-driven systems been evaluated?
- **RQ4:** How do MR platforms reciprocally drive and contribute to GenAI development?

Through this structured analysis, we articulate a unified understanding of the **evolving symbiotic relationship** between visual GenAI and MR, while identifying critical gaps within existing frameworks and delineating promising avenues for future research on adaptive, human-centered interaction design.

II. BACKGROUND: OVERVIEW OF AI-DRIVEN VISUAL CONTENT CREATION

GenAI facilitates the creation of diverse digital assets (e.g., images, videos, and 3D models) that are essential for immersive MR experiences. This section examines core techniques that balance high-fidelity content generation with the computational efficiency required for real-time interaction.

Foundational breakthroughs, including Generative Adversarial Networks (GANs) [54] and the Transformer architecture [55], paved the way for modern GenAI. Transformers, capable of training on vast unlabeled datasets, enabled the rise of LLMs like OpenAI’s GPT-4 [56]–[58]. These models demonstrate near-human performance in summarization, translation, and domain-specific reasoning, delivering operational benefits through automated content creation and personalized communication [14], [59], [60].

Recent research focuses on optimizing autoregressive architectures [61]–[63] for efficiency and context length. For instance, DeepSeek [64] utilizes a Mixture-of-Experts architecture to selectively activate parameters during inference; Mamba [65] and Jamba [66] employ State Space Models

for linear scaling with context length. In VR/AR, these advancements are pivotal for powering intelligent agents. LLMs enable dynamic Non-Player Characters (NPCs) [67], virtual storytelling assistants [68], [69], and adaptive educators [70] to maintain coherent, context-aware dialogue, enhancing the intuitiveness of virtual worlds (further discussed in Section IV).

A. Image and Video Generation

Advancements in 2D generative models have revolutionized image synthesis and editing. While GANs [54], [72] and variational autoencoders (VAEs) [73] laid the groundwork, diffusion models [7], [74], [75] have become the dominant paradigm for high-fidelity generation (Figure 2). Tools like DALL-E 3 [76] and Stable Diffusion [71] excel at text-to-image synthesis, while Midjourney [77] is favored for artistic stylization. Beyond generation, models such as Instruct-Pix2Pix [78] and commercial tools like Adobe Firefly [79] integrate generative editing directly into creative workflows. For further comprehensive reviews of image generation, readers can refer to Bie et al. [80] and Li et al. [81].

Video generation extends these capabilities by addressing temporal coherence and scene dynamics. Platforms like Runway (Gen-2/3) [82], Pika Labs [83], and Stability AI’s Stable Video Diffusion [84] generate moving visuals from text or image inputs. More advanced models, such as OpenAI’s Sora [85], produce long-form, high-fidelity clips with complex character motion, while SEINE [86] focuses on smooth transitions for storytelling. For a comprehensive discussion on video generation models, we direct readers to Melnik et al. [87] and Xing et al. [88].

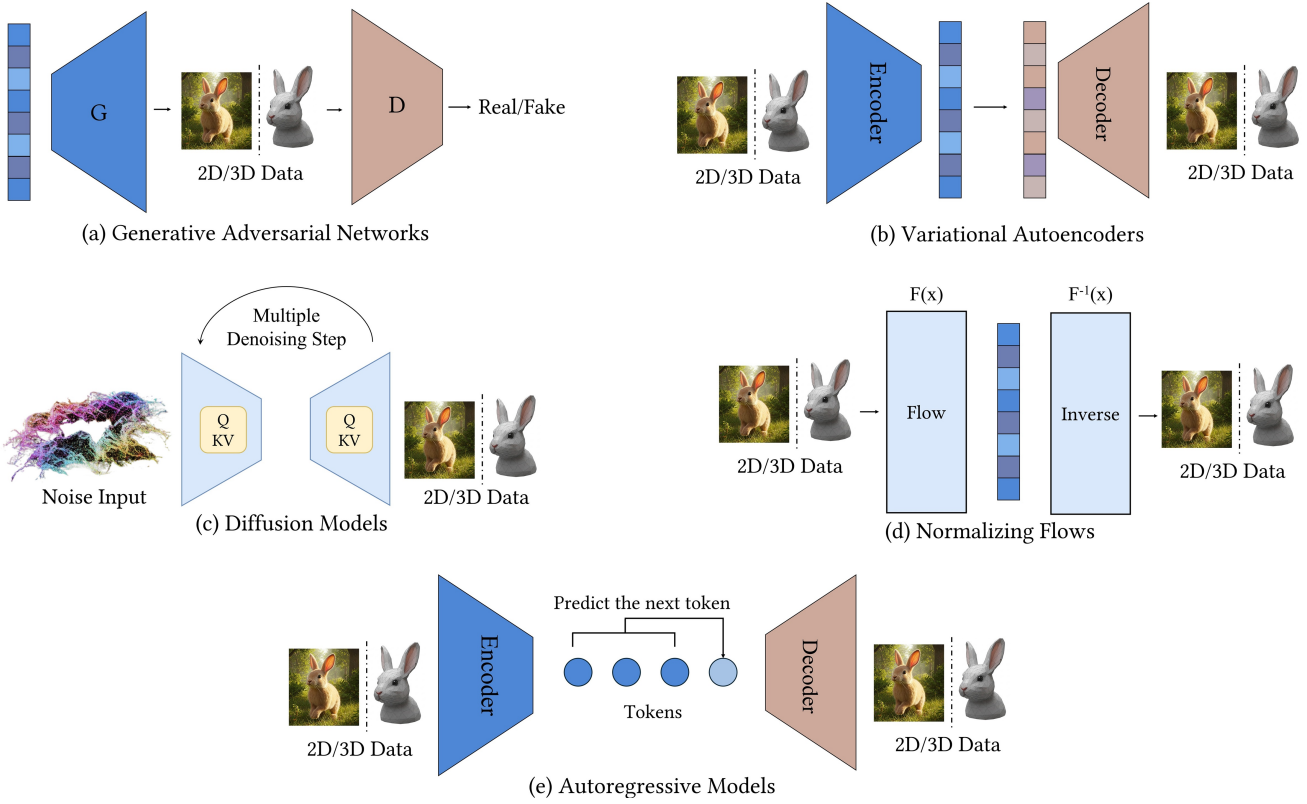


Fig. 2. Exemplary 2D/3D generation models. These models generally require a large amount of data (e.g., billion-level for 2D [71], and million-level for 3D [20]) to support diverse content creation.

B. 3D Generation

Generating 3D content is inherently more complex than 2D due to the requirements for volumetric and multi-view coherence. As detailed in Li et al. [37] and Xia et al. [89], recent generation approaches rely on neural representations, ranging from Neural Radiance Fields (NeRF) [90], differentiable point clouds [91], voxel grids [92], polygonal meshes [93], Neural Signed Distance Functions [94] to 3D Gaussian Splatting (3DGS) [95]. The choice of representation often dictates the strengths and weaknesses of subsequent generation process (e.g., compatibility with graphics pipeline [96], [97], stylization [98], [99], and generation speed [100]–[102]).

Two primary paradigms dominate 3D generation. **Optimization-based methods** (e.g., DreamGaussian [103], ProlificDreamer [104]) use Score Distillation Sampling (SDS) [105] to distill 3D geometry from pre-trained 2D diffusion models. While they leverage strong 2D priors, per-object optimization can be slow. Conversely, **feedforward methods** (e.g., GANFusion [106], DiffTF++ [107]) generate 3D representations in a single inference pass. These are faster but often struggle to match the quality of optimization-based approaches. Recent unified frameworks aim to bridge this gap. For example, TRELLIS [20] employs a Structured 3D Latent representation trained via Rectified Flow [108], enabling high-quality generation adaptable to various formats (i.e., NeRF, 3DGS, Mesh). These advancements [109]–[112] are crucial for MR, streamlining the creation of assets for gaming, simulations [113], and virtual try-ons [114], thereby

democratizing 3D content creation [115], [116]. For readers seeking a more comprehensive examination of 3D generation advancements, the surveys by Xie et al. [117], Shi et al. [118], and Li et al. [119] offer valuable in-depth reviews.

III. METHODOLOGY

This review charts the development of visual GenAI for interaction enhancement in MR, focusing on literature published between 2018 and 2025. Guided by the PRISMA method [51], we followed structured phases of identification, screening, and eligibility assessment to assemble a highly relevant and focused body of work. Beyond quantitative aggregation, we qualitatively synthesize the selected studies around two core dimensions: (1) the design, applications, challenges, and evaluation of GenAI–MR integration; and (2) the reciprocal role of immersive interactions in advancing GenAI techniques.

A. Identification

The initial phase of our review involves a systematic identification of relevant scientific papers. This process was designed to capture high-quality research from prominent publication venues and keyword searches across databases.

First of all, to identify key publication venues, we consulted Google Scholar Metrics [53] to select influential conferences and journals in the fields of Computer Graphics (CG), Computer Vision (CV), and Human-Computer Interaction (HCI). Our search targeted leading venues such as **CHI**, **IMWUT**, **UIST**, **IUI**, **TOCHI**, **DIS**, **VRST**, **SIGGRAPH**,

TABLE II
BOOLEAN INSTRUCTIONS FOR ACM DIGITAL LIBRARY, IEEE XPLORE, SPRINGER, AND SCIENCEDIRECT.

Database	Boolean Instructions
ACM Digital Library	Title:((Generative AI OR Visual Content Generation) AND (Virtual Reality OR Augmented Reality) OR (Immersive Interaction)) Abstract:((Generative AI OR Visual Content Generation) AND (Virtual Reality OR Augmented Reality) OR (Immersive Interaction))
IEEE Xplore	Abstract: (Generative AI OR Visual Content Generation) AND (Virtual Reality OR Augmented Reality) AND (Interaction*)
Springer ScienceDirect	Title: (Interaction* OR Generation*) AND (Virtual Reality OR VR OR Augmented Reality OR AR) Abstract: (Generative AI OR Content Generation) AND (Virtual Reality OR Augmented Reality OR Mixed Reality) AND (Interaction*)

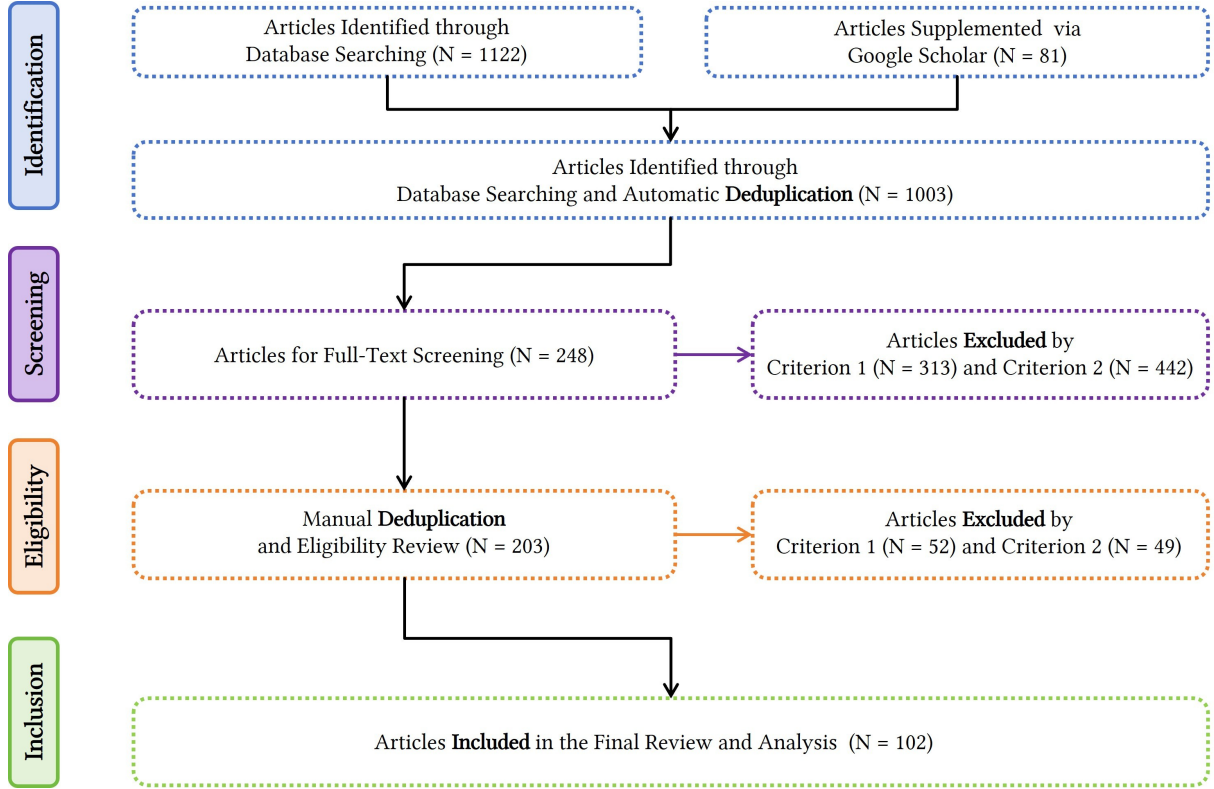


Fig. 3. Literature search and selection process. We follow the PRISMA [51] procedure, comprising four phases: identification, screening, eligibility, and inclusion. Each phase details the number of records included and excluded.

TOG, CVPR, ICCV, ECCV, IJCV, WACV, NIPS, TVCG, IJHCI, IEEE VR, AIxVR, IJVR, ISMAR, and Frontiers in Virtual Reality. Papers from these venues were primarily retrieved from four major scientific databases: the ACM Digital Library, IEEE Xplore, ScienceDirect, and SpringerLink. To ensure comprehensive coverage and minimize omissions, as suggested by Du et al. [120], we performed a supplementary keyword search on Google Scholar. Recognizing the rapid advancements in GenAI and MR, we also included critical technical reports and preprints from arXiv to capture the latest developments.

Our search strategy employed a combination of keywords to retrieve articles at the intersection of our defined themes. These keywords were grouped into two main categories:

- **Core Concepts.** “immersive interaction*,” “generative

AI*,” and “visual content generation*” were used to capture articles discussing MR interactions influenced by or related to GenAI and content creation. The asterisk (*) served as a wildcard to include variations such as “generation” “generating,” and “generative,” and similarly for “interaction.”

- **Medium Specificity.** “augmented reality,” “virtual reality,” “mixed reality”, “AR,” “VR” and “MR” were included to ensure that the selected studies were situated within VR/AR/MR. These terms were searched within the title or abstract of the articles. Table II details search queries for the four aforementioned databases.

The inclusion criteria for articles were as follows: (a) publication date between January 2018 and October 2025; and (b) publication type. Works published after 2025 were discussed

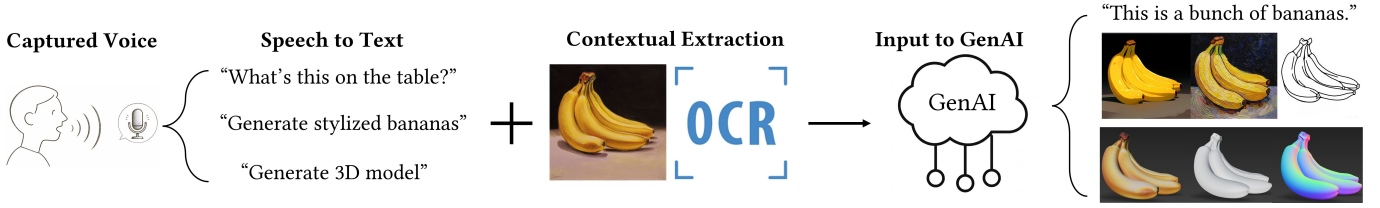


Fig. 4. A general pipeline in multimodal interactions by integrating GenAI-powered generation and reasoning. Starting from the user’s speech signal or direct text inputs, the GenAI’s feedback is subsequently visualized to the user or processed further within integrated systems (e.g., sMoRe [121], DreamRelive [122]).

only as emerging examples and are not counted in the 102-paper corpus. We included full-text research papers, works-in-progress, and posters. This inclusive approach to publication type aimed to build a comprehensive collection, covering diverse aspects and emerging applications of visual content generation in MR. While shorter formats (e.g., workshop) may not consistently meet the same evaluative rigor as full papers, their inclusion was deemed valuable for capturing the latest explorations by researchers leveraging GenAI and multi-modal interactions in MR.

B. Screening and Eligibility

To ensure the reliability and consistency of the screening process, the selection was conducted by two researchers who are active in both MR and GenAI. Both reviewers independently screened the titles and abstracts of a random sample of 10% of the retrieved papers to establish inter-rater reliability. Disagreements were resolved through discussion to refine the inclusion criteria before proceeding to the full dataset. Following the abstract screening, the full texts were assessed. In cases where the eligibility of a paper was ambiguous, a third senior researcher was consulted to reach a consensus, ensuring that only articles contributing substantive empirical or technical insights into the intersection of GenAI and MR interaction were included. In sum, we screened whether the above-identified papers met the following **two criteria**.

The first main selection criterion was defined to identify studies examining visual GenAI for diverse interaction enhancements in MR. Articles were included if they investigated the integration of visual GenAI to address interaction challenges or enrich UI/UX within immersive media, particularly those involving diverse input and output modalities to create seamless and dynamic experiences [120], [123]. Conversely, studies were excluded if their primary focus was on: (1) non-visual GenAI techniques (e.g., audio generation [124], motion generation [125], natural language processing without a primary visual interactive component [126], or MR interactive enhancements with reconstruction instead of generation techniques [127]); (2) generative methods not readily compatible with, or optimized for, real-time MR rendering pipelines (e.g., computationally intensive 4D content generation [128], [129]); or (3) the enhancement of UI/UX in MR using AI techniques other than GenAI (e.g., relying solely on object detection and tracking [24], [130], or facial recognition [131]).

Another central focus of this review is the dynamic interplay between advancements in visual GenAI and the evolution

of MR platforms, moving beyond direct applications to investigate MR’s role as advanced simulation environments that actively shape generative model development (**RQ4**). Our selection criteria emphasize studies exploring: (1) the use of MR for immersive analytics and “user-in-the-loop” paradigms for the iterative refinement of visual GenAI systems, as exemplified by research in interactive neural network visualization within VR [132] or accessible machine learning (ML) for immersive interaction [133]; and (2) the application of MR to generate high-quality, context-rich synthetic data for training and enhancing visual GenAI, such as through photorealistic virtual environments for data generation (e.g., UnrealROX [134]) or MR-based data labeling [135]. As shown in Figure 3, 442 papers were excluded in this stage. For example, UnrealCV [136] and Kubric [137] utilize 3D game engines for synthetic image generation, annotation, and model testing to improve vision-oriented tasks. However, these works do not use MR interactions, immersive user input, or spatial feedback to train, steer, evaluate, or refine visual GenAI models. Therefore, RQ4 characterizes the mechanisms observed among eligible MR-for-GenAI studies, rather than implying that MR platforms inevitably contribute to GenAI development. This focus enables a comprehensive analysis of MR’s dual capacity as a medium for GenAI experiences and a contributor to the GenAI lifecycle. Consequently, **203** articles were retained during these stages.

C. Data Analysis

We employed a systematic data extraction rubric and an integrated coding strategy combining deductive and inductive approaches to determine and analyze the final set of **102** research works (Figure 1).

Data Extraction. Two authors independently extracted data from the selected articles using a standardized form capturing: (1) MR Modality (VR, AR); (2) Generative Model Architecture (e.g., GANs, Diffusion, LLMs, VLMs); (3) Interaction Modality (e.g., gaze, gesture); and (4) Evaluation Metrics (usability, performance, latency). To further support quantitative synthesis, we computed cross-tabulations in Figure 6.

Analysis Procedure. We applied a deductive approach to categorize our findings into the predefined RQs (e.g., Spatial Creation, Intelligent Collaborators). As shown in Fig. 5, we adopted a multi-label tagging scheme. This strategy reflects the hybrid nature of GenAI–MR systems, many of which combine content generation, interaction guidance, and authoring support. Subsequently, we utilized an inductive approach to allow novel themes to emerge from the data, specifically regarding

the reciprocal role of MR as a testbed for GenAI. We also utilized Affinity Diagramming [138] to group extracted codes into higher-level themes. Weekly meetings were held among the research team over a two-month period to discuss emerging clusters, resolve coding discrepancies, and iteratively refine the taxonomy until saturation was achieved. This rigorous process ensured that findings accurately reflected the current state of the art while minimizing subjective bias.

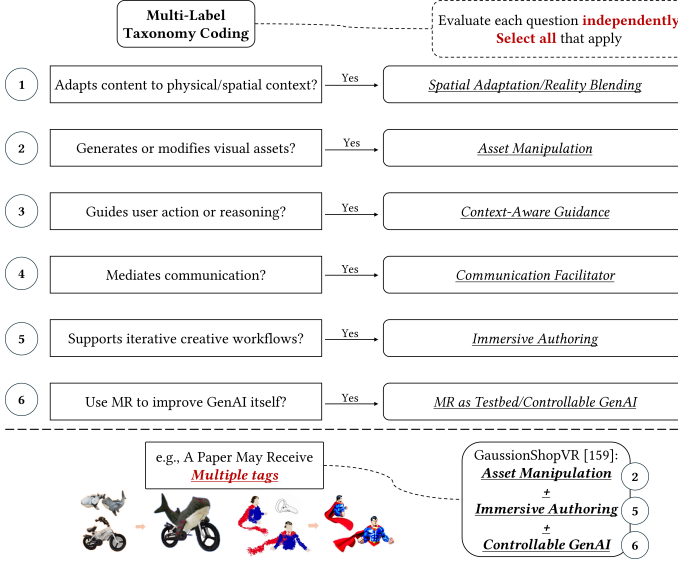


Fig. 5. Decision flowchart for multi-label taxonomy coding. Each reviewed paper was evaluated against all coding questions corresponding to the proposed research categorizations.

IV. RESULTS AND DISCUSSIONS

Following the reviewing structure of Hirzle et al. [31], we present a typology of the state-of-the-art research at the intersection of MR and visual GenAI. Our analysis addresses following RQs through corresponding thematic investigations: (1) the role of GenAI in spatial interaction and dynamic scene understanding (**RQ1**), (2) the deployment of GenAI as Intelligent Agents in MR (**RQ2**), (3) evaluation metrics and approaches for GenAI-driven immersive systems (**RQ3**), and (4) the utilization of MR technologies to advance GenAI models, including synthetic data generation, immersive visualization, fine-grained controllable generation, and model fine-tuning (**RQ4**). Each section synthesizes findings from representative works, identifying emerging interaction paradigms and methodological innovations. Figure 6 complements the thematic taxonomy with a cross-tabulated synthesis of model paradigms, interaction modalities, and evaluation practices. It shows that LLM-based orchestration and VLM/MLLM-based visual reasoning are most common in context-aware guidance, communication facilitation, and immersive authoring, whereas explicit 3D generation is concentrated in asset manipulation and authoring (Figure 6(a)). It also shows a task-dependent division of interaction modalities: speech/text dominates asset manipulation, guidance, communication, and authoring; gesture, hand input, and sketching are especially visible in authoring; and scene capture is central to spatial adaptation, reality

blending, and MR-for-GenAI (Figure 6(b)). Evaluation practices remain uneven, with interviews and satisfaction ratings appearing more often than standardized usability, workload, latency, or generation-quality metrics (Figure 6(c)). These patterns indicate that the field has advanced beyond interactive prototypes but may still need shared benchmarks to assess generative quality, spatial controllability, responsiveness, and user agency.

A. GenAI for Spatial Creation and Scene Understanding

As categorized in Table III, this section examines how GenAI transforms immersive interactions through Spatial Arrangement, Reality Blending, and Asset Manipulation. Figure 4 demonstrates a general interaction pipeline.

Spatial Arrangement and Adaptation. GenAI advances MR interfaces by enabling content that dynamically responds to user actions and environmental contexts [144]. LLMs and VLMs function as coordination layers, bridging multimodal perception with generative adaptation to facilitate seamless transitions between virtual and physical environments [143].

For object-level manipulation, systems such as XR-Objects [123] integrate real-time tracking [168], [169] with Multimodal LLMs [14] to render context-aware menus, transforming physical items into intelligent augmented objects. Similarly, PaperToPlace [140] spatializes traditional instructions by anchoring procedural steps to relevant physical objects, thereby reducing the need for context switching. Beyond object-centric augmentation, generative reasoning extends to situationally aware layout optimization. While earlier methods relied on geometry or predefined semantics [170], [171], SituationAdapt [139] employs VLM reasoning to dynamically adjust UI layouts and prevent occlusion based on social and environmental factors. This illustrates a clear trajectory in adaptive MR: evolving from static geometry-based optimization to task-aware arrangements, and finally toward the socially intelligent orchestration of spatial content [141].

Reality Blending. A core aspiration of MR is the unified integration of real and virtual elements. GenAI achieves this by dynamically manipulating digital content to align with users' environmental perception. At the object level, VR-Props [151] predicts deformation parameters, ensuring virtual objects mirror their physical counterparts. TransforMR [150] advances this concept via video inpainting [172], enabling virtual elements to replace physical objects while preserving their contextual properties.

Furthermore, GenAI-driven inpainting [173]–[176] enables sophisticated alteration of real-world captures. Headset-MR [148], [153] leverages these techniques to allow interactive manipulation of deformable 3DGS scenes. To optimize interactions within these blended environments, SIM2VR [152] integrates RL models [177] to create feedback loops between user simulation and VR applications. Furthermore, blending extends beyond the visual; DeepTaste [154] utilizes GANs for food-to-food translation, demonstrating GenAI's capacity to modify multisensory perceptions. While avatar generation [178], [179] enhances immersion, it constitutes a distinct domain reviewed comprehensively by Wang et al. [180] and is outside the scope of environmental augmentation.

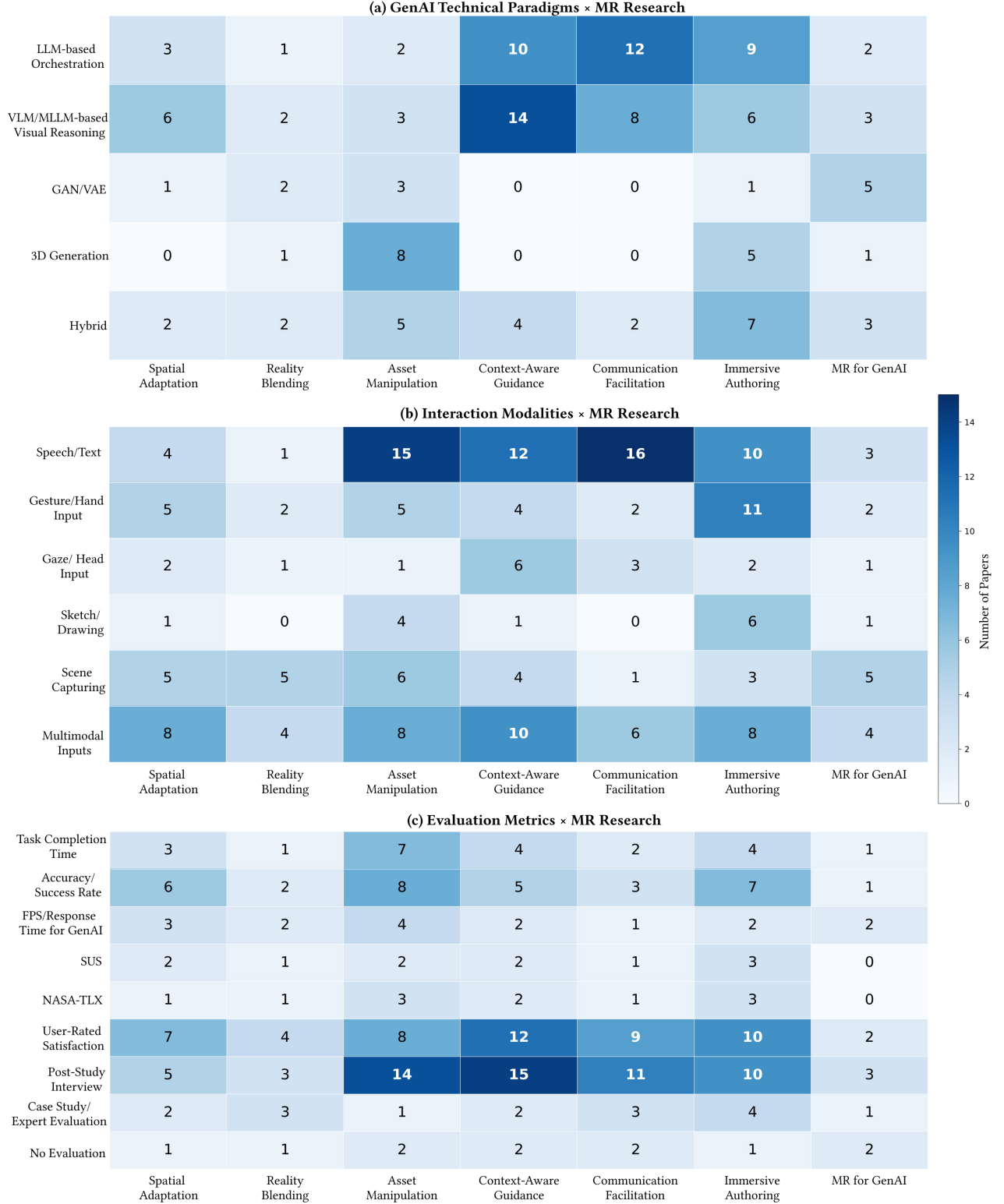


Fig. 6. Cross-tabulation of visual GenAI in MR. We summarize the relationships among (a) GenAI technical paradigms, (b) interaction modalities, (c) evaluation metrics, and MR research areas with a multi-label coding scheme. Each cell reports the number of papers tagged with both row and column labels.

Asset Generation and Manipulation. GenAI democratizes spatial creation by automating the transition from intuitive inputs (speech, gaze, gesture) to 2D and 3D content [160]–[162]. Frameworks like sMoRe [121] and Behravan et al. [155] unify LLM-based command interpretation

with text-to-3D APIs [181], allowing users to generate and refine assets through natural language and layout maps. In VR, MineVRA [156] and DreamRelive [122] utilize adaptive generation models [182], [183] to produce contextually immersive experiences, such as transforming spoken narratives

TABLE III
REFERENCES OF GENAI FOR SPATIAL CREATION AND SCENE UNDERSTANDING.

Category	Number of Papers	Peer-Reviewed	Preprint	Extended Abstract/Poster/Workshop	Representative Works
Spatial Adaptation	10	7 (70%)	1 (10%)	2 (20%)	[123], [139]–[147]
Reality Blending	7	6 (85.7%)	0 (0.0%)	1 (14.3%)	[148]–[154]
Asset Manipulation	18	11 (61.1%)	5 (27.8%)	2 (11.1%)	[30], [121], [122], [143], [144], [155]–[167]

TABLE IV
REFERENCES OF GENAI AS INTELLIGENT COLLABORATORS.

Category	Number of Papers	Peer-Reviewed	Preprint	Extended Abstract/Poster/Workshop	Representative Works
Context-Aware Guidance	24	17 (70.8%)	4 (16.7%)	3 (12.5%)	[29], [184]–[193] [194]–[206]
Communication Facilitator	19	13 (68.4%)	3 (15.8%)	3 (15.8%)	[120], [185], [190], [205], [207]–[212] [163], [197], [198], [201], [213]–[217]
Immersive Authoring	15	12 (80%)	1 (6.7%)	2 (13.3%)	[30], [142], [159], [162], [214], [218]–[227]

into manipulable visual particles.

Beyond static geometry, GenAI facilitates the definition of dynamic behaviors. Voice2Action [157] establishes a pipeline to categorize LLM-powered actions for environment configuration, while DreamCodeVR [158] employs GPT-4 [56] as an interactive coding assistant. By interpreting spoken commands to generate and execute C# code, these systems enable users to iteratively sculpt not just the appearance of MR experiences, but their programmatic logic and interactivity [144], [163].

B. GenAI as Intelligent Collaborators

We then examine the deployment of GenAI as intelligent collaborators in MR, evolving beyond static tools to provide context-aware guidance, communication support, and creative partnership (Table IV).

Context-Aware Guidance. GenAI demonstrates advanced capabilities in interpreting user intentions and environmental cues to deliver dynamic, intelligent guidance [200], [228]. In VR, recent systems leverage VLMs to enhance spatial awareness. For example, VR-GPT [229] enables voice-based task guidance without intrusive visual overlays, while VRSearch [186] and VOI-VR [187] utilize VLMs to translate natural language queries into precise 3D object identification, effectively mitigating the limitations of traditional rule-based systems in occluded environments [230].

Collaborative capabilities are further enriched through multimodal understanding. Systems like PANDALens [188] and VirtuWander [29] synthesize diverse inputs—including audio, gaze patterns, and spatiotemporal data—to generate coherent travel narratives and multimodal responses that combine visual annotations with spatial audio. In AR, this contextual awareness extends to spatial disambiguation and accessibility. ARTiST [184] enables rapid target identification through contextual specification, while WorldScribe [189] provides real-time scene descriptions for visually impaired users. Furthermore, the convergence of GenAI’s contextual capabilities with MR simulations [231] has catalyzed advances in professional training, positioning GenAI as integral to next-generation systems across domains such as medical diagnosis [185], [198], construction [194], [195], robot control [196].

Communication Facilitator. GenAI functions as an intermediary to facilitate adaptive, context-sensitive, and socially

meaningful interactions. In immersive learning, it acts as a bridge between learners and knowledge domains. Zhu et al. [211] utilize GenAI to empower avatars with adaptive role-switching capabilities for history education, while VisionARy [212] integrates GPT-4 [56] and YOLO-v7 [232] to ground language learning in real-world surroundings. These applications personalize dialogue to sustain engagement and manage cognitive load [205], [216].

Beyond education, GenAI enhances social presence and fluency. In intergenerational communication, AI-powered probes transform abstract topics into interactive artifacts, balancing dialogue dynamics between users [120]. This extends to the setup of NPCs, where GPT-driven architectures enable virtual agents to exhibit scene-aware behaviors and unscripted interactions [207], [208]. By embedding user identity models [233] and perspective-aware frameworks [209], these systems shift NPCs from static entities to dynamic interlocutors capable of adapting to users’ evolving cognitive states and social contexts.

Immersive Authoring. GenAI transforms immersive authoring by supporting mixed-initiative co-creation [214]. Users can articulate creative intentions via speech, gestures, and sketches, enabling iterative workflows [223], [224]. Tools like VRCopilot [234] and MS2Mesh-XR [221] leverage these modalities for 3D asset and layout design, while ImagineAR [142] empowers non-experts to design outdoor AR experiences via speech and real-time scene understanding [235].

For complex scene manipulation, systems such as Dreamcrafter [30] (Figure 7(a)) and GaussianShopVR [159] (Figure 7(b)) employ text-driven prompts to edit NeRF and 3DGS representations, allowing for precise control over object generation and stylization. This extends to dynamic media, where Imagine360 [236] and VideoCraft [149] (Figure 7(c)) synthesize 360-degree content and blend physical and virtual elements through spatial-layer editing. At the prototyping stage, tools like ProtoDreamer [218] and FusionProtor [219] merge physical modeling with generative pipelines, transitioning low-fidelity tangible models into high-fidelity digital assets. Furthermore, to address computational latency in generative workflows, systems like Calliope [222] utilize MR’s spatial affordances to parallelize design exploration, transforming wait times into opportunities for collaborative evaluation.



Fig. 7. Examples of immersive authoring workflows: (a) integration of text- and image-driven 3D editing in MR; (b) and (c) utilization of 3D representations to enhance creative spatial controls. Images were from the above-cited works under CC-BY 4.0 license or with permissions from the authors.

C. Evaluating GenAI-Driven MR Systems

Evaluating integrated systems of GenAI and MR requires balancing task efficiency, system performance, and user experience. As summarized and categorized in Table V, unlike traditional MR evaluations, where interaction techniques are often benchmarked by accuracy or usability, GenAI-driven systems introduce additional dimensions such as model responsiveness, generative quality, and integration overhead to capture both human performance and GenAI capabilities. Due to the absence of standardized evaluation frameworks for these emerging systems, **post-study interviews** (61/102) have become predominant for deriving design principles and capturing subjective user experiences [139], [216], [226]. Figure 8 summarizes the distribution of general evaluations adopted by reviewed works.

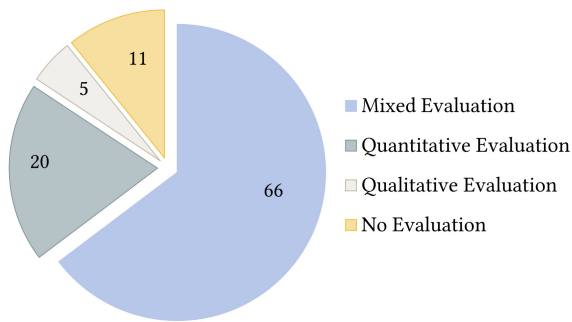


Fig. 8. Visualization of evaluations applied in GenAI-Driven MR systems.

Task Performance. Quantitative metrics remain central when GenAI is deployed for task guidance and immersive authoring. **Task Completion Time** is widely applied, showing whether GenAI streamlines creation and accelerates MR interactions [123], [140], [165], [210]. System overhead, such as GPU consumption [155] and FPS [148], [150], [151], [153], is also considered when evaluating whether generative processes scale without compromising responsiveness, particularly critical for reality-blending scenarios, where visual continuity underpins immersion. **Accuracy Metrics** [158], [186], [191], [228], [229], [242] further capture how GenAI affects the precision of user interactions with virtual objects or navigation tasks, revealing the reliability of these integrated systems. To probe interaction dynamics and fluidity more deeply, some studies also introduce **Action Count** (the number

of GenAI interventions per task) [156], [157], [194], [210], [228] and **Time-to-Response** (latency of GenAI output) [143], [221], [222], [237], [238]. These metrics highlight GenAI as a creative and intelligent collaborator that enhances interaction efficiency and immersion. Besides, standard generation metrics (e.g., Structural Similarity Index Measure and CLIP Score) are also employed to quantify the quality of the generated assets within MR [145], [202], [215].

System Usability and Cognitive Load. Beyond raw performance, usability metrics highlight how well GenAI integrates into immersive workflows. Standardized evaluations such as the **System Usability Scale (SUS)** [121], [140], [155], [156], [159] and **NASA-TLX** [142], [191], [194], [234] are frequently applied to capture usability, satisfaction, and workload. Complementing these scales, post-study interviews provide richer insights into whether GenAI strengthens or disrupts user agency and creative flow [123], [139], [165]. **User-Rated Satisfaction** [29], [212] further provides direct feedback on system acceptance, cognitive load, and user engagement. For instance, Pei et al. [243] report that GenAI reduced cognitive burden in assembly tasks through this rating, while Situation-Adapt [139] demonstrates that users found GenAI-empowered layouts more contextually appropriate and less intrusive. In sum, these evaluations provide a nuanced account of users' immersion in GenAI-enhanced MR experiences [237], [248].

Case Study and Expert Evaluation. To contextualize findings of the proposed interaction techniques, many works adopt controlled lab studies [139], [149], case-based demonstrations [161], [200], or expert evaluations [162], [213]. For instance, GenAI-controlled NPCs [207] have been evaluated through a scenario-driven case study that highlights both potential and limitations of dynamic character behaviors toward MR users. Similarly, scene generation pipelines employ visual case studies to demonstrate diversity and immersion, often supported by side-by-side visual comparisons with baseline methods [30], [148], [153]. Expert evaluations add another important layer of assessment: designers often assess generative prototyping tools for fidelity, while novice users evaluate integrated systems for feasibility [197], [213]. For example, Du et al. [120] engage experts and families to assess how AI-supported VR fostered intergenerational communications. These evaluations go beyond conventional metrics, grounding results in **domain-specific** effectiveness and social impact.

Evaluation Patterns and Gaps. Beyond metric cate-

TABLE V
REFERENCES OF EVALUATION METRICS AND PRACTICES.

Category	Number of Papers	Peer-Reviewed	Preprint	Extended Abstract/Poster/Workshop	Example Citations
Action Count of GenAI	15	12 (80.0%)	2 (13.3%)	1 (6.7%)	[120], [156], [157], [184], [188], [194], [210], [228] [30], [143], [162], [204], [207], [216], [237]
Time to Response from GenAI	11	7 (63.6%)	2 (18.2%)	2 (18.2%)	[143], [153], [155], [158], [197], [201], [213], [221], [222], [237], [238]
Task Completion Time	22	17 (77.3%)	5 (22.7%)	0 (0%)	[123], [140], [155]–[158], [165], [187], [210], [229] [145], [162], [164], [186], [191]–[193], [203], [218], [227], [228], [239]
Accuracy Metrics	32	23 (71.9%)	6 (18.8%)	3 (9.3%)	[140], [158], [165], [184], [186], [189], [191], [228], [229], [240] [145], [152], [159], [192], [196]–[198], [215], [226], [227], [241] [135], [160], [162], [193], [200], [223], [225], [237], [239], [242], [243] [142], [148], [150], [151], [153]
Frame Per Second (FPS)	5	4 (80.0%)	0 (0%)	1 (20.0%)	
Case Study	9	7 (77.8%)	2 (22.2%)	0 (0.0%)	[120], [161], [200], [203], [207], [215], [216], [243], [244] [121], [123], [139], [140], [154]–[156], [165], [210], [211]
Post-Study Interview	61	48 (78.7%)	8 (13.1%)	5 (8.2%)	[122], [158], [162], [193], [202], [204], [212], [214]–[216], [228], [229] [29], [167], [184], [186], [188], [189], [191], [208], [241], [245] [120], [148], [150], [163], [166], [194], [217], [226], [246] [30], [142], [159], [218]–[220], [222], [236], [242], [247] [133], [143], [147], [164], [192], [203], [213], [223], [225], [248] [29], [121]–[123], [139], [154], [156], [158], [211], [212] [147], [163], [164], [167], [184], [187], [188], [208], [215], [228], [229] [30], [142], [148]–[150], [159], [194], [219], [234], [236] [162], [200], [202], [205], [213], [216], [224], [237], [247], [248] [166], [192], [193], [203], [204], [217], [225]–[227], [244], [246] [121], [140], [148], [155], [156], [159], [162], [164], [167], [200], [218] [121], [140], [142], [156], [158], [162], [164], [191], [194], [227], [234] [145], [148], [151], [215], [221], [223]
System Usability Scale (SUS)	11	8 (72.7%)	3 (27.3%)	0 (0%)	
NASA Task Load Index (TLX)	11	7 (63.6%)	4 (36.4%)	0 (0%)	
Subjective Visual Comparison	6	6 (100%)	0 (0%)	0 (0%)	

gories, the reviewed evaluation practice reveals a field that is still largely prototype-oriented. Most evaluations combined quantitative and qualitative evidence, while a smaller number relied exclusively on quantitative benchmarks, qualitative feedback, or no evaluation; interviews and user-rated satisfaction were substantially more common than standardized usability, workload, latency, or generation-quality measures (Figures 6(c) and 8). Among empirical studies reporting participant numbers, sample sizes are typically modest, often ranging from 8 to 20 participants. This pattern is reflected across recent GenAI–MR investigations: XR-Objects [123] was evaluated with 8 participants; LLMR [165] involved 11 participants; PaperToPlace [140] used two within-subject cohorts with 12 participants each; SituationAdapt [139] conducted a 12-participant in-lab study; VirtuWander [29] and PANDALens [188] included 12 and 16 participants respectively; RemVerse [210] was assessed with 14 older adults; MineVRA [156] and ImagineAR [142] each involved 20 participants. The dominant design was a short, single-session controlled comparison, often within-subject and counterbalanced, where users completed object manipulation, task guidance, or immersive authoring tasks under a GenAI–MR condition and one or more baselines. Exploratory authoring, reality-blending, and experience-oriented systems often relied on expert reviews, case studies, or visual comparisons when objective ground truth was difficult to define. This evaluation setup is appropriate for demonstrating feasibility, but it may limit the strength of claims about learning effects, transfer across contexts, and robustness in everyday MR use.

The more consequential gap is not simply a lack of metrics, but a narrow evaluation horizon. Current studies privilege immediate efficiency, perceived usability, and short-term generation quality, while comparatively few examine longitudinal engagement, retention of benefits, calibrated trust, recovery from erroneous generations, or whether users can understand, contest, and reverse AI-driven spatial changes. Accessibility and inclusive designs also remain unevenly addressed. Systems

such as WorldScribe [189] and RemVerse [210] show promising directions for low-vision users and older adults, yet accessibility is often treated as an application-specific contribution rather than a routine evaluation dimension across GenAI–MR systems. Future evaluations should report prior experience, accessibility needs, and cultural or linguistic diversity, and test systems across heterogeneous environments and failure scenarios. We suggest a mature evaluation framework may combine task performance and system latency with controllability, reversibility of operations, perceived agency, long-term usage reliance, and accessibility across diverse user groups.

D. MR Environments as Testbeds for GenAI Development

MR platforms function not only as consumers of GenAI content but also as controlled testbeds for its development. By offering controlled environments for synthetic data generation and spatial manipulation, MR facilitates the systematic fine-tuning of models across diverse tasks. This establishes a reciprocal relationship where spatial computing actively shapes AI evolution. For example, Pei et al. [243] demonstrate that fine-tuning LLMs with synthetic, MR-derived dialogue significantly improves assembly training, illustrating how immersive media can directly enhance generative performance. We categorize this role into the following two primary mechanisms.

Synthetic Data Generation. The advancement of visual GenAI is often bottlenecked by the scarcity of large-scale, well-annotated datasets. MR platforms offer a scalable solution by generating high-fidelity, task-specific training data under controlled conditions. This approach addresses data limitations not only for GenAI but also for broader ML tasks, including object classification, detection, tracking, RL, and depth estimation [249], [250]. MR-based data generation operates through three key modalities:

- **Simulation Frameworks:** Integrated systems like the Unity ML-Agents Toolkit [251] and UnrealROX [134] (Figure 9(a)) leverage game engines to construct sophis-

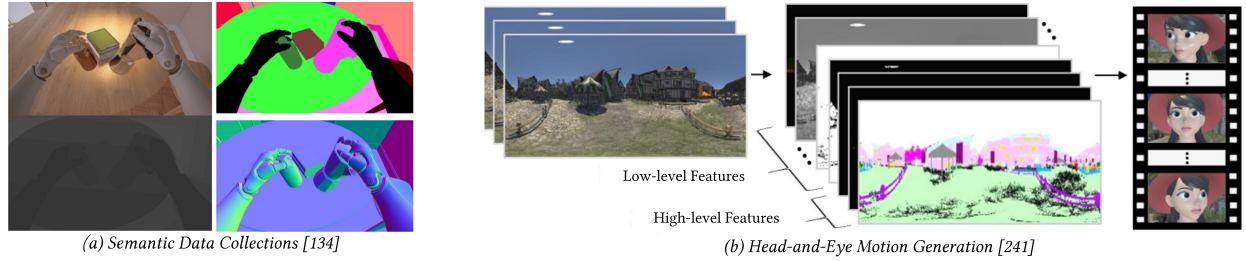


Fig. 9. Examples of synthetic generation via immersive media: (a) first-person-view (FPV) hand interaction data in diverse indoor scenarios and (b) Context-Aware Head-and-Eye Motion Dataset (CAHE). Images were from the above-cited works with permissions from copyright holders.

ticated environments. These frameworks facilitate embodied learning and automate annotation pipelines, creating data that would be costly to capture in the real world.

- **Human-Centric Capture:** Recent approaches utilize MR headsets to record naturalistic spatial interactions. These hybrid methods bridge the reality gap by combining human behaviors with scalable annotation to advance robot learning [252] and VLMs [253].
- **Generative Augmentation:** Synthetic pipelines can be further enriched via generative models. Techniques involving GANs [240], [249] or diffusion models [241], [254], [255] augment VR-generated data to improve diversity and context awareness, capturing subtle behavioral cues essential for human-centered AI.

In summary, MR ecosystems offer a versatile pipeline that combines high-fidelity simulation, human-in-the-loop collection, and generative augmentation. This effectively mitigates data scarcity and narrows the sim-to-real gap for complex spatial modeling (Figure 9(b)).

MR Input for Controllable GenAI. The convergence of user-in-the-loop methodologies and multimodal interaction in MR enhances the steerability of GenAI. Unlike desktop interfaces, immersive environments allow users to guide model behavior, curate data, and manipulate outputs through embodied actions. Direct manipulation enables rapid feedback loops. Systems like ALive [132] allow users to adjust neural parameters via gestures, while InteractML [133] provides a node-based VR interface for recording and pruning models. These tools democratize the training process, offering richer 3D manipulations than standard visualizations [245], [256].

Immersive tools improve the quality of training data. LabellingFlora [135] demonstrates that VR-based segmentation allows both novices and experts to produce accurate 3D annotations. Similarly, FusionProtor [219] integrates physical prototypes with AI-driven generation, enabling iterative refinement through mixed physical-virtual interactions.

AR systems introduce semantics and spatial reasoning into the generation pipeline. Frameworks such as FocalPoint [239], GaussianShopVR [159], and VideoCraft [149] utilize fine-grained spatial inputs to anchor GenAI outputs to human intent. Furthermore, Xu et al. [238] exemplify this coordination by allowing users to search and track objects via voice commands while simultaneously steering model behaviors. These systems highlight immersive media as a coordination layer between human intent and generative adaptation.

V. FUTURE RESEARCH OPPORTUNITIES

As the intertwined fields of GenAI and MR continue to evolve, numerous promising research directions remain to be explored. This section outlines key areas where future work can further enhance the synergetic development of GenAI in MR (Figure 10(a)-(f) respectively). For example, DrawTalking [257] shows that sketch-and-speech interaction can support code-free construction of interactive worlds, but also reveals a key limitation for current GenAI-enhanced creative workflows: LLM-based command translation may be too slow for instant feedback and can reduce user control and determinism.

A. Improving Generation Quality, Efficiency, Editing Granularity, and Contextual Reasoning

Firstly, we suggest that future research should continue to focus on improving the quality and efficiency of visual GenAI to ensure seamless real-time interaction experiences. The current limitations in computational resources often force a trade-off between generation quality and response speed, particularly in mobile MR devices with limited computing power. Research should pivot toward model quantization techniques that allow generative inference to occur closer to the user without sacrificing the realism essential for immersion.

3D Generation Enhancement. Recent advances, such as Clay [12] and TripoSR [102], demonstrate promising capabilities in creating 3D assets from single images within seconds. Future work could explore more efficient generative models and neural representations that can produce detailed and textured 3D models with minimal generation latency.

In parallel, a highly promising research direction involves making generated assets more directly **compatible with design pipelines**. For instance, part-aware generation [258] enables the decomposition of shapes into parts, facilitating smoother editing and composability. Recent explorations into explicit mesh generation (e.g., TextDeformer [96], Text2Mesh [97], and HeadEvolver [259]) have empowered designers by preserving 3D attributes such as UV coordinates and blendshapes. These approaches streamline the creative process, as they allow for rigging and animation to be easily transferred across multiple generated assets, minimizing manual post-processing. Complementary lines of work (e.g., Text2VDM [260] and GenVDM [261]) generate artist-friendly sub-assets such as vector displacement maps and sculpting brushes that attach to surfaces, further reducing manual detailing work. Continued advancement in these areas will pave

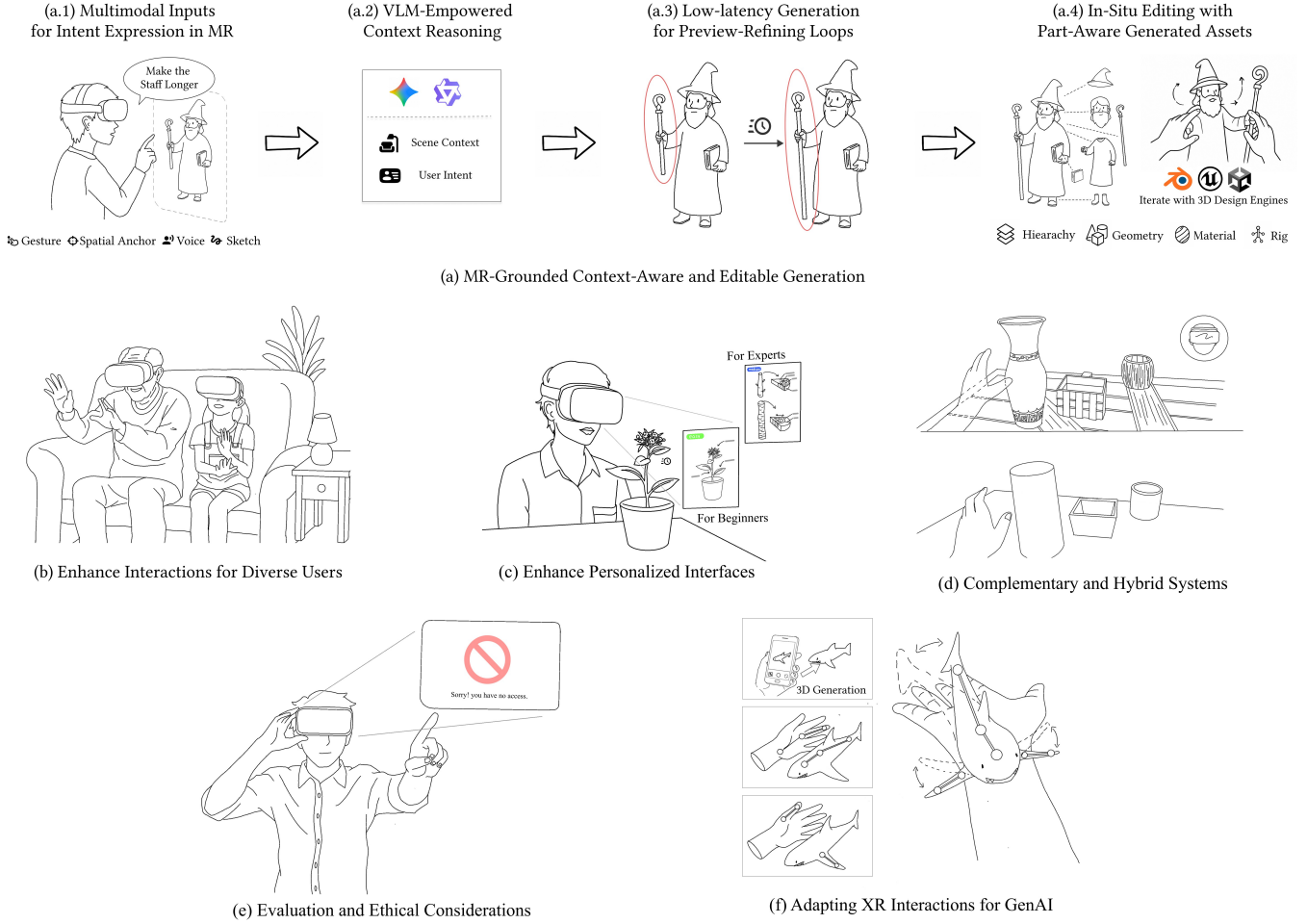


Fig. 10. Future research directions. (a)–(f) summarize the concepts in Sections V-A through V-F, respectively.

the way for powerful generation systems that are not only computationally efficient but also seamlessly integrated into the toolchains of creative professionals.

Context-Aware Multi-modal VLM Responses. VLMs used in MR would benefit from more refined abilities to access and synthesize accurate information from verifiable sources [189]. For example, scene understanding in MR can be further enhanced if VLMs can quickly respond with more accurate semantically labeled 3D bounding boxes for each object [262]. This enhancement would enable more contextually relevant and factually grounded spatial interactions. Future systems should integrate historical information [263] and multimodal inputs—such as gaze fixation, hand gestures, and voice intonation—to provide long-term, context-aware, and adaptive responses. For example, a system should understand that a user pointing at a virtual chair while saying “make this look older” requires specific texture degradation on that specific mesh, necessitating tight synchronization between low-level vision understanding (e.g., YOLO [232]), VLMs, and 3D generation pipelines.

B. Enhance Interactions for Diverse Users

As we have discussed, Du et al. [120], and Wang et al. [213] have begun exploring how GenAI can facilitate

communications between various age groups in MR. Further research could continue to investigate how GenAI bridges cultural and generational gaps, making MR more accessible and meaningful across demographic boundaries and device constraints [264]. Likewise, Tokmurziyev et al. [265] have proposed that LLM-driven glasses with haptic feedback can assist in navigation for visually impaired individuals. Future research could extend these capabilities to provide a richer environmental understanding and more intuitive multi-modal feedback systems by harnessing the high-quality environmental simulation capabilities of MR.

GenAI also opens powerful avenues for crafting multi-modal narratives within creative domains. In cultural heritage, for instance, GenAI presents significant opportunities to digitize traditional arts to preserve and share cultural practices among diverse audiences (e.g., Centennial Drama Reimagined [266]). However, achieving seamless and authentic contextual storytelling is challenging. Current implementations, such as mural recreations by Zhao et al. [167], reveal persistent issues with scene-level consistency and the faithful reproduction of historical details. For creative applications to be both inspiring and respectful, future development must refine GenAI frameworks that carefully balance imaginative engagement with cultural fidelity.

C. Toward Adaptive and Autonomous Personalized Interfaces

Advancing personalized interfaces in MR through GenAI constitutes a key direction for future research. Prior work emphasizes that personalization is central to making MR experiences both usable and meaningful [139]. For example, MineXR [267] demonstrates how users benefit from decomposing applications into modular widgets, creating layouts that reflect not only individual needs but also contextual and task-specific requirements. Their findings indicate that users often prefer functionality-based interfaces over application-based ones, emphasizing direct access to core features (e.g., email inbox, recipe steps) while minimizing unnecessary navigation. Similarly, studies on adaptive MR interfaces [268] highlight the importance of context-aware adaptation that responds to changes in environments (e.g., office, kitchen, public spaces), activities (e.g., work, leisure, sports), and dynamic states (e.g., object-manipulating scenarios).

GenAI provides new opportunities to extend these approaches by enabling adaptive, generative, and mixed-initiative personalization. Unlike rule-based or predefined adaptive systems, GenAI models can synthesize novel layouts, styles, or multimodal representations that better align with individual preferences. By leveraging multimodal signals such as gaze [269]–[273], gestures [274], environmental cues [275], or even prior interaction histories [276], [277], these systems may continuously infer user intent and generate situated assistance that evolves with the task context. This shift also changes the role of the user: GenAI-MR systems may act not only as passive tools, but also proactively adapt spatial content. Therefore, a critical future direction is to conceptualize GenAI-MR interaction through human–GenAI autonomy levels [278], which may range from:

- **Level 0: User-Driven.** Users manually configure all spatial interface components (e.g., 3D object creation [156] and geometry recomposing [279]).
- **Level 1: AI-Assisted.** GenAI provides recommendations (e.g., alternative layouts [139] and organization [280]) but relies on users’ decisions and confirmation.
- **Level 2: Mixed-Initiative.** Users and GenAI jointly shape interface behavior, with generative models modifying non-critical elements autonomously while keeping the user in the loop (e.g., adaptive learning [281], UI Agent [282], and cooking assistance [123], [205]).
- **Level 3: AI-Directed Personalization.** GenAI autonomously orchestrates all interface configurations based on long-term patterns, shifting the user toward oversight rather than direct manipulation (e.g., working memory management [204] and attention modeling [193]).
- **Level 4: Fully Autonomous Personalization.** MR environments continuously reconstruct themselves to optimize user experience without explicit user input, raising substantial questions around transparency, agency, and safety (e.g., proactive organization of everyday moments [206], [283] and always-on MR Agent [284]).

These autonomy levels can be understood as a GenAI-MR extension of established automation taxonomies [285], where automation varies not only by degree but also by

function, including information acquisition, analysis, decision making, and action implementation. Recent research also highlights an emerging tension: while increased autonomy can reduce effort and enhance fluidity, over-automation risks diminishing user agency, obscuring system logic, or fostering over-reliance on AI-generated interpretations. As generative models increasingly participate in spatial reasoning [139], content arrangement [156], and even social mediation inside MR [120], concerns arise regarding cognitive atrophy, persuasive interface shaping, and the reinforcement of biased behavioral inferences [216].

Future work should move beyond treating autonomy as a design feature and evaluate it as a dynamic human–GenAI relationship in MR [279], [285], [286]. Across different autonomy levels, MR studies should examine not only efficiency and usability, but also whether users maintain situational awareness, calibrated trust, creative ownership, and the ability to understand, contest, override, or reverse AI-driven changes [278], [287]. Within immersive media, evaluation protocols should also include automation reliability, intervention costs, recovery from erroneous generations, provenance comprehension, and users’ perceived agency under both normal and failure conditions [288], [289]. Such evidence can guide GenAI-MR systems toward adaptive assistance that reduces interaction burden while preserving human judgment, ensuring that autonomy scaffolds rather than replaces user creativity.

D. Complementary and Hybrid Systems

While GenAI provides strong creative capabilities, the full richness of human interaction in immersive environments stems from hybrid systems that unify diverse AI techniques with specialized hardware. PANDALens [188] and VirtuWander [29], for example, use multimodal signals to support LLM-based content generation and guidance. Our reviews suggest the contextual grounding and adaptability of GenAI-enabled MR systems can be further enhanced by incorporating sensory input, physical interaction, and user feedback.

Research in sensor fusion and computer vision has demonstrated how perception-driven techniques enhance spatial understanding. For example, DepthLab [24] illustrates how computer vision pipelines facilitate depth-aware AR interactions, including occlusion handling and surface-sensitive content placement. When paired with GenAI, these methods could allow generated content to become dynamically grounded in the geometry and semantics of the surrounding environment.

Equally critical are tangible interfaces that bridge digital and material spaces. Systems such as PaperToPlace [140] and HoloDoc [25] demonstrate the potential of physical artifacts to anchor MR interactions in ways that are intuitive and embodied. Extending these approaches with GenAI could enable workflows where sketches, documents, or everyday objects are enriched with dynamically generated content, or where physical manipulation seamlessly blends with generative augmentation. The future of MR will not be defined by GenAI alone but by its integration with complementary systems of sensing, perception, and tangibility. Such hybrid approaches promise immersive experiences that are both personalized and grounded in the richness of multi-modal human activity.

E. Evaluation and Ethical Considerations

While many of these evaluation metrics overlap with those used for traditional non-MR applications, the dynamic and often autonomous nature of GenAI within such spatial contexts introduces unique considerations. The interaction is no longer solely with a predefined system but with an evolving learning entity that shapes the virtual world and offers dynamic responses [46]. Therefore, a critical avenue for future research lies in refining evaluation frameworks to address the ethical dimensions of GenAI in spatial computing. Methodologies must transcend traditional performance metrics such as efficiency and fidelity of generated content to encompass human-centric factors like long-term engagement, perceived agency, and trust. Furthermore, evaluations should scrutinize content bias, model transparency, and the social implications of multimodal interactions (e.g., copresence [290], [291] and Interpersonal Trust Scale [292]). Equally important is ensuring user autonomy [278] in persuasive or adaptive GenAI and safeguarding against misleading generated responses.

Privacy and Data Protection. As immersive GenAI systems increasingly rely on detailed contextual information, questions of privacy and data governance become increasingly important. Balancing personalization with security and access control [293] requires careful design of data flows and storage. Systems that continuously monitor users' environments, behaviors, or physiological signals must minimize exposure of sensitive information, for instance, by performing on-device processing or transmitting only task-relevant statistical features [294]. Similar challenges are evident in collecting data for personalized interaction designs [295]. Approaches such as local computation, selective use of proxy signals, or privacy-preserving transformations can mitigate risks while supporting adaptive, user-centered experiences.

F. Spatial MR Interactions for Controllable GenAI

Advanced visual GenAI models increasingly support fine-grained control, but MR systems still expect interaction techniques that can translate embodied user actions into model constraints. In this context, 3DUI should be viewed not only as a way to manipulate generated assets, but also as a way to express user intent before, during, and after generation. Existing systems such as VideoCraft [149] and GaussianShopVR [159] have demonstrated how passthrough views, spatial perception, object selection, and layer-based editing can anchor generative processes in the users' surrounding scenes. Future work can build on established 3D object selection techniques [239], [296]–[298] and 3D layer-based interactions [299] to let users specify visual attributes, object parts, spatial hierarchy, viewpoints, motion, and editing scope with greater precision.

A more concrete direction is to develop proxy-guided MR interactions for controllable GenAI. Instead of relying on text or image prompts, users could steer models through intermediate MR proxies, including selected objects, segmentation masks, hand poses, spatial anchors, object parts, scene layers, motion paths, and semantic intent cues. For instance, GeneratedReality conditions interactive video generation on tracked

head and joint-level hand poses [300]; HOICraft maps object-part affordances and hand movements into VR interaction behaviors [301]; and PlayWrite converts embodied storytelling actions into editable intent frames and story marbles [302]. These works show that MR interactions can provide spatial, temporal, behavioral, and semantic proxies that help users specify not only what should be generated, but also where it should appear, how its parts should relate, how it should transform, and how it should remain editable across iterations.

Future work should formalize these proxies as reusable interaction and model-conditioning primitives. Such primitives should be inspectable, reversible, and composable, allowing users to lock constraints, revise generated results, and recover from failed interpretations without restarting the workflow. Evaluation should therefore move beyond visual fidelity and compare proxy-guided workflows with prompt-based workflows on controllability, user agency, spatial accuracy, cognitive load, and revision effort. In this framing, we envision future MR systems as a distinctive interaction layer that makes GenAI more controllable within situated, iterative, and human-centered design workflows.

VI. LIMITATIONS

Although this survey follows a PRISMA-guided structured review process, several limitations should be noted. First, our corpus may be affected by search, venue, and timing bias. We focused on leading venues in HCI, computer graphics, computer vision, GenAI, and immersive media, and supplemented database retrieval with Google Scholar and arXiv searches to reduce omissions. This strategy captures a broad and timely body of work, but it cannot guarantee exhaustive coverage, especially because GenAI–MR research is evolving quickly, and relevant systems may first appear as preprints, posters, or open-source projects [303]–[305]. Borderline cases further illustrate this limitation: Handle-based mesh deformation guided by multi-view VLM predictions is highly relevant but does not primarily study MR interaction [306]; MiXR explores in-situ compositional modeling [279], but its publication timing and fit with our review categories make it more suitable for future updates. Following Stefanidi et al. [307], who emphasize the importance of clear terminology and methodological transparency in HCI literature reviews, our survey should be interpreted as a structured snapshot of the rapidly evolving field rather than as an exhaustive or definitive account.

Second, our taxonomy and quantitative synthesis are subject to coding and interpretation bias. GenAI–MR research often combines multiple technical paradigms, interaction modalities, and application goals. A single system may involve scene understanding, asset manipulation, immersive authoring, and user-in-the-loop refinement [149], [156], [159]. Although we used PRISMA, multi-label coding, iterative discussion, and consensus-based resolution, some category boundaries remain interpretive. Other researchers might reasonably organize the same corpus by model architecture, MR device, application domain, or evaluation method. Therefore, our taxonomy should be understood as one organizing lens for this growing field, not as a fixed classification. Similarly, the evaluation trends

reported in our survey describe how GenAI–MR systems have been assessed, not which design paradigms are consistently superior. Future reviews may conduct finer-grained analyses of evaluation quality, longitudinal use, reproducibility, and participant diversity as the field matures.

Finally, our survey focuses on visual GenAI in MR. We excluded or de-emphasized work focused primarily on non-visual generation, general AI-based MR perception, reconstruction-only techniques, or systems without a substantive MR interaction component. This scope keeps the review focused, but it may underrepresent adjacent advances in audio, haptics, robotics, and embodied AI that increasingly shape GenAI–MR workflows. As model capabilities, commercial tools, and MR hardware continue to evolve, future surveys should revisit the corpus, update the taxonomy, and examine whether multimodal systems require new categories.

VII. CONCLUSION

This survey underscores the transformative potential of visual Generative AI in immersive media, where the synergy between GenAI and MR promises to revolutionize multimodal spatial interaction, content creation, and human-AI collaboration. By systematically mapping existing research across techniques, interaction frameworks, and evaluation methods, we highlight both the progress achieved and the gaps that persist at this emerging intersection. The findings highlight significant opportunities for advancing adaptive, human-centered interaction designs, refining model integration for spatial computing, and leveraging immersive platforms to accelerate the development of GenAI. To conclude, this work lays the groundwork for future research that aims not only to address current challenges but also to unlock the full potential of visual GenAI as a catalyst for more naturalistic, scalable, and intelligent immersive experiences.

AUTHOR CONTRIBUTION STATEMENT

Duotun Wang led the project conception and design, coordinated research activities, performed the primary data collection and theme-based analysis, drafted the figures, and the original manuscript. Linjie Qiu contributed to the data analysis, validated the systematic review methodology, and provided critical revisions to the manuscript to enhance its writing quality. Linlin Cai designed the visualizations, specifically the survey structure and teaser, and critically revised the manuscript to enhance the discussion depth. Yue Jiang provided methodological guidance, contributed to data analysis and interpretation, and critically revised the manuscript to enhance innovative contributions. Mingming Fan and Zeyu Wang supervised the project, acquired funding, contributed to the research planning and study conception, participated in data analysis discussions, and critically revised the manuscript. All authors reviewed and approved the final version of the manuscript and agree to be accountable for all aspects of the work, ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

DATA AVAILABILITY STATEMENT

This article is a survey study and did not generate new experimental data. Searched and reviewed papers are from research databases, including IEEE Xplore, ACM Digital Library, Springer, ScienceDirect, and arXiv. As a survey paper, the authors confirm that the data supporting the findings of this study are available within the article itself.

COMPETING INTERESTS STATEMENT

The authors declare that they have no known competing interests that could have appeared to influence this paper.

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